

LESSON 5

BOOK CHAPTERS 24

ELECTRIC POTENTIAL

Problem 21 (Book Chapter 24)

The ammonia molecule NH_3 has a permanent electric dipole moment equal to 1.47 D, where $1D = 1\text{debye unit} = 3.34 \times 10^{-30} C.m$. Calculate the electric potential due to an ammonia molecule at a point 52.0 nm away along the axis of the dipole. (Set $V = 0$ at infinity.)

Answer:

We have

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{p \cos \theta}{r^2} \right)$$

Given

$$p = 1.47 D = 1.47 \times 3.34 \times 10^{-30} C - m$$

$$r = 52 \text{ nm} = 52 \times 10^{-9} m$$

$$\theta = 0^\circ \text{ and } \cos 0^\circ = 1 \quad V = ?$$

$$V = \frac{9 \times 10^9 \times 1.47 \times 3.34 \times 10^{-30}}{(52 \times 10^{-9})^2} = 16.34 \times 10^{-6} \text{ Volt}$$

Potential Gradient

Example 23.13 Potential and field of a point charge

From Eq. (23.14) the potential at a radial distance r from a point charge q is $V = q/4\pi\epsilon_0 r$. Find the vector electric field from this expression for V .

SOLUTION

IDENTIFY and SET UP: This problem uses the general relationship between the electric potential as a function of position and the electric-field vector. By symmetry, the electric field here has only a radial component E_r . We use Eq. (23.23) to find this component.

EXECUTE: From Eq. (23.23),

$$E_r = -\frac{\partial V}{\partial r} = -\frac{\partial}{\partial r}\left(\frac{1}{4\pi\epsilon_0} \frac{q}{r}\right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

so the vector electric field is

$$\vec{E} = \hat{r}E_r = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

EVALUATE: Our result agrees with Eq. (21.7), as it must.

An alternative approach is to ignore the radial symmetry, write the radial distance as $r = \sqrt{x^2 + y^2 + z^2}$, and take the derivatives of V with respect to x , y , and z as in Eq. (23.20). We find

$$\begin{aligned}\frac{\partial V}{\partial x} &= \frac{\partial}{\partial x}\left(\frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{x^2 + y^2 + z^2}}\right) = -\frac{1}{4\pi\epsilon_0} \frac{qx}{(x^2 + y^2 + z^2)^{3/2}} \\ &= -\frac{qx}{4\pi\epsilon_0 r^3}\end{aligned}$$

and similarly

$$\frac{\partial V}{\partial y} = -\frac{qy}{4\pi\epsilon_0 r^3} \quad \frac{\partial V}{\partial z} = -\frac{qz}{4\pi\epsilon_0 r^3}$$

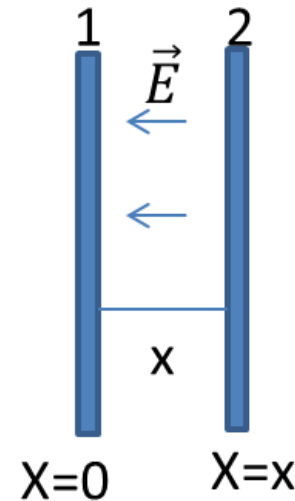
Then from Eq. (23.20),

$$\begin{aligned}\vec{E} &= -\left[\hat{i}\left(-\frac{qx}{4\pi\epsilon_0 r^3}\right) + \hat{j}\left(-\frac{qy}{4\pi\epsilon_0 r^3}\right) + \hat{k}\left(-\frac{qz}{4\pi\epsilon_0 r^3}\right)\right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{r}\right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}\end{aligned}$$

This approach gives us the same answer, but with more effort. Clearly it's best to exploit the symmetry of the charge distribution whenever possible.

Problem 36 (Book Chapter 24)

The electric potential V in the space between two flat parallel plates 1 and 2 is given (in volts) by $V = 1500x^2$, where x (in meters) is the perpendicular distance from plate 1. At $x = 1.3 \text{ cm}$, (a) what is the magnitude of the electric field and (b) is the field directed toward or away from plate 1?



(a) We have

$$E_x = -\frac{\partial V}{\partial x} = -\frac{\partial}{\partial x}(1500x^2) = -3000x$$

$$E_x = -3000 \times 0.013 = -39 \frac{\text{V}}{\text{m}}$$

$$\vec{E} = E_x \hat{i} = 39(-\hat{i})$$

Magnitude of $\vec{E}$ is

$$E = 39 \frac{\text{V}}{\text{m}}$$

Given

$$V = 1500x^2$$

At $x = 1.3 \text{ cm} = 0.013 \text{ m}$,

$$E = ?$$

The direction of electric field,

$$\vec{E} = ?$$

(b) The direction of electric field is toward plate 1, because $\vec{E} = 39(-\hat{i}) \frac{\text{V}}{\text{m}}$

Problem 37 (Book Chapter 24)

What is the magnitude of the electric field at the point $(3.00\hat{i} - 2.00\hat{j} + 4.00\hat{k})\text{ m}$ if the electric potential in the region is given by $V = 2.00xyz^2$, where V is in volts and coordinates x , y , and z are in meters?

Answer:

We know

$$E_x = -\frac{\partial V}{\partial x} = -\frac{\partial}{\partial x}(2xyz^2) = -2yz^2 = -(2)(-2)(4^2) = 64 \frac{\text{V}}{\text{m}}$$

$$E_y = -\frac{\partial V}{\partial y} = -\frac{\partial}{\partial y}(2xyz^2) = -2xz^2 = -(2)(3)(4^2) = -96 \frac{\text{V}}{\text{m}}$$

$$E_z = -\frac{\partial V}{\partial z} = -\frac{\partial}{\partial z}(2xyz^2) = -4xyz = -(4)(3)(-2)(4) = 96 \frac{\text{V}}{\text{m}}$$

$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k} = 64\hat{i} - 96\hat{j} + 96\hat{k}$$

Therefore,

$$|\vec{E}| = \sqrt{(64)^2 + (-96)^2 + (96)^2} = 150.09 \frac{\text{V}}{\text{m}}$$

Given

$$V = 2xyz^2$$

And

$$(x, y, z) = (3, -2, 4)$$

$$|\vec{E}| = ?$$

Do by yourself .

- I. CALCULATE THE NET ELECTRIC POTENTIAL AT POINT P [ASSUME $V = 0$ AT INFINITY] (FIG. I)
- II. CALCULATE THE NET ELECTRIC POTENTIAL AT POINT MIDWAY BETWEEN THE CHARGES [ASSUME $V = 0$ AT INFINITY]
- III. CALCULATE THE NET ELECTRIC POTENTIAL AT THE CENTER OF THE SQUARE IF THE LENGTH OF EACH SIDE OF THE SQUARE IS 5.0 CM [ASSUME $V = 0$ AT INFINITY] .
- IV. CALCULATE THE NET ELECTRIC POTENTIAL AT THE CENTER OF THE SQUARE IF THE LENGTH OF EACH SIDE OF THE SQUARE IS 8.0 CM. [ASSUME $V = 0$ AT INFINITY]

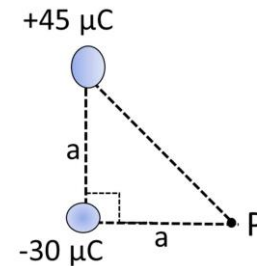


Fig. I

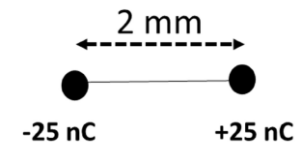


Fig II

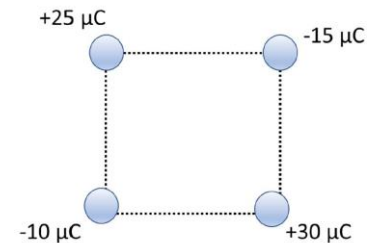


Fig. III

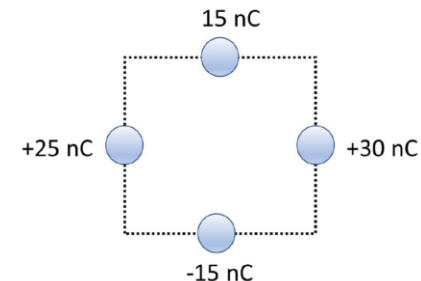


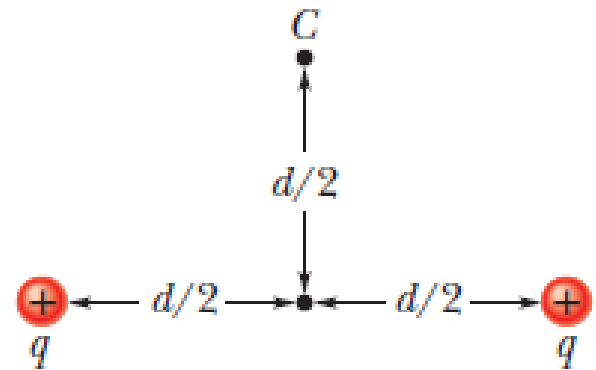
Fig. IV

Let's try !!!

Problem 35: The electric potential at points in an xy plane is given by $V = 2.0 \left(\frac{\text{V}}{\text{m}^2} \right) x^2 - 3.0 \left(\frac{\text{V}}{\text{m}^2} \right) y^2$. In unit-vector notation, what is the electric field at the point (3.0 m, 2.0 m)?

Problem 47: What is the *escape speed* for an electron initially at rest on the surface of a sphere with a radius of 1.0 cm and a uniformly distributed charge of $1.6 \cdot 10^{-15} \text{ C}$? That is, what initial speed must the electron have in order to reach an infinite distance from the sphere and have zero kinetic energy when it gets there?

Problem 88: Two charges $q = 2.0 \text{ } \mu\text{C}$ are fixed a distance $d = 2.0 \text{ cm}$ apart (adjacent Fig.). (a) With $V = 0$ at infinity, what is the electric potential at point C? (b) You bring a third charge $q = 2.0 \text{ } \mu\text{C}$ from infinity to C. How much work must you do? (c) What is the potential energy U of the three-charge configuration when the third charge is in place?





Thank You
